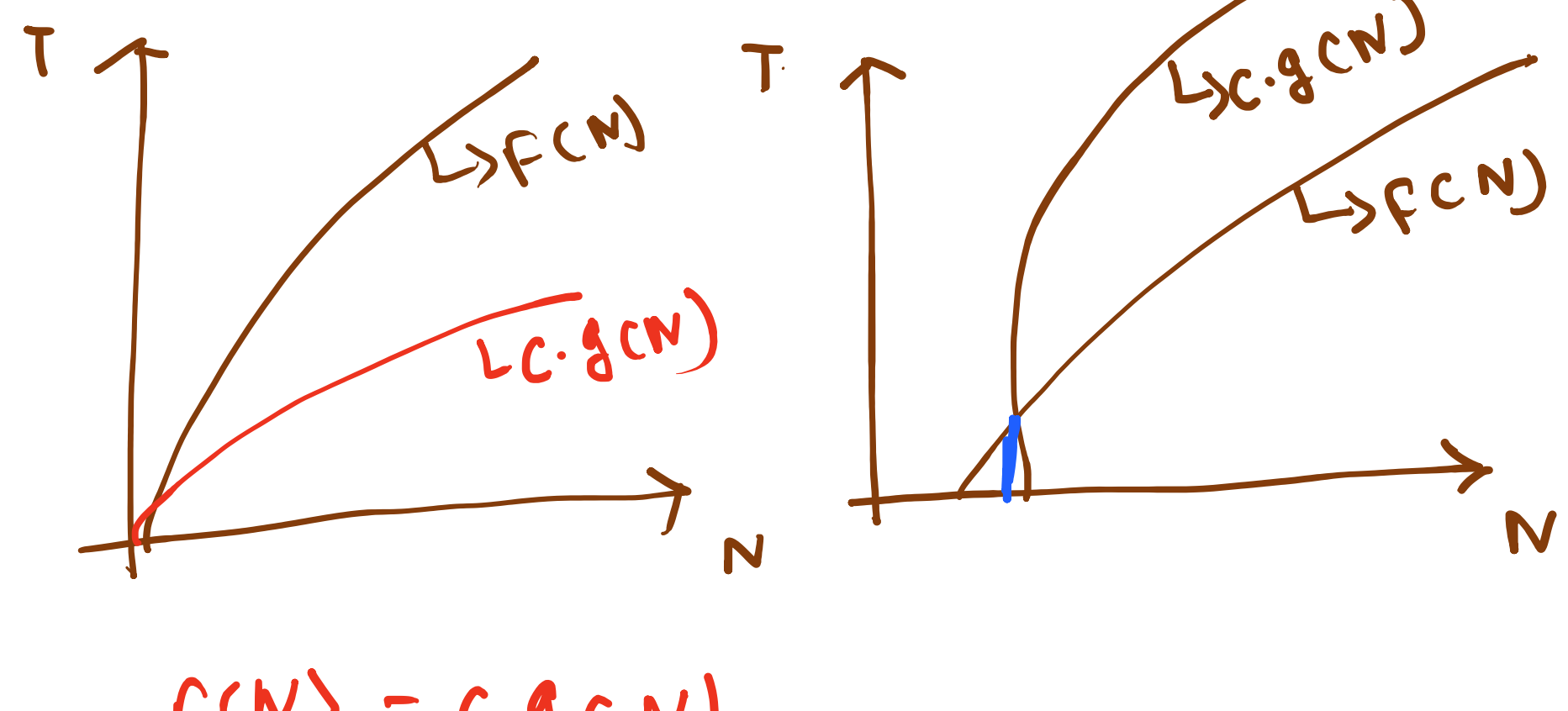


Time & Space Complexity

asymptotic notation vs Experimental Way



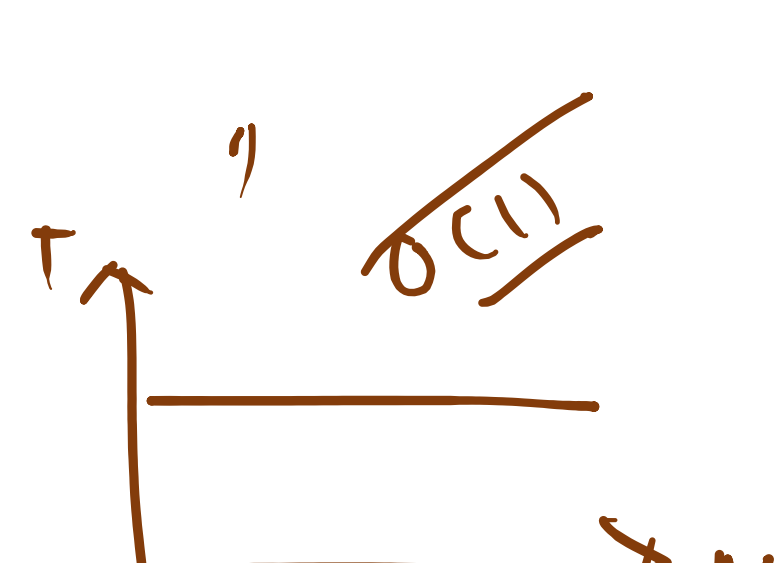
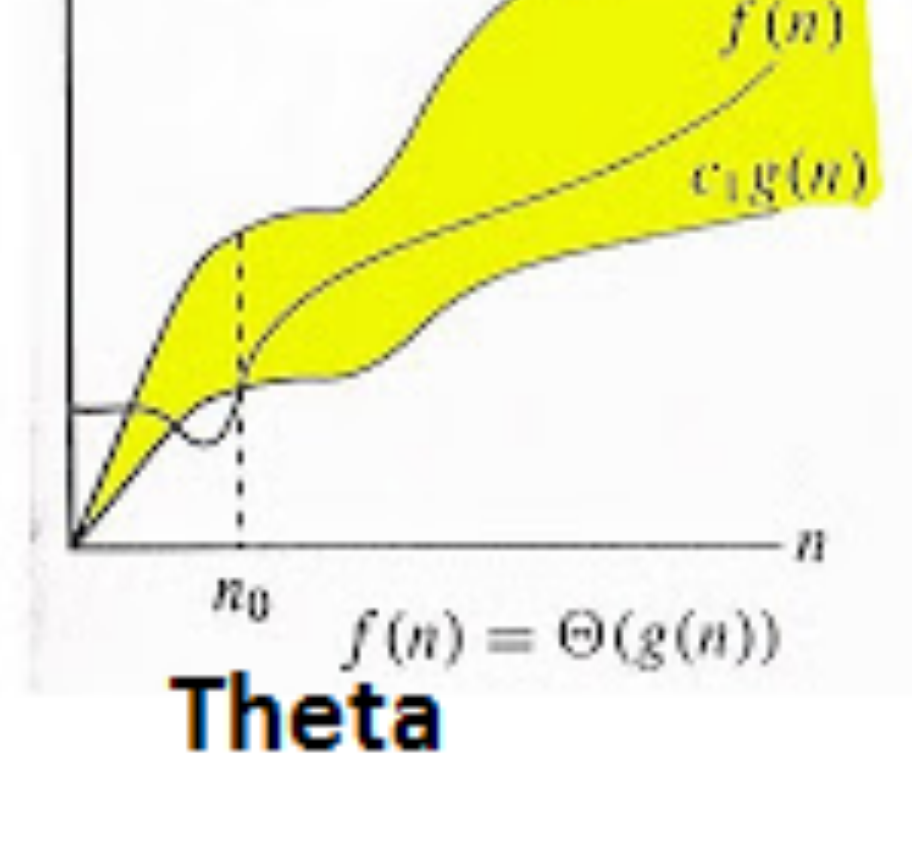
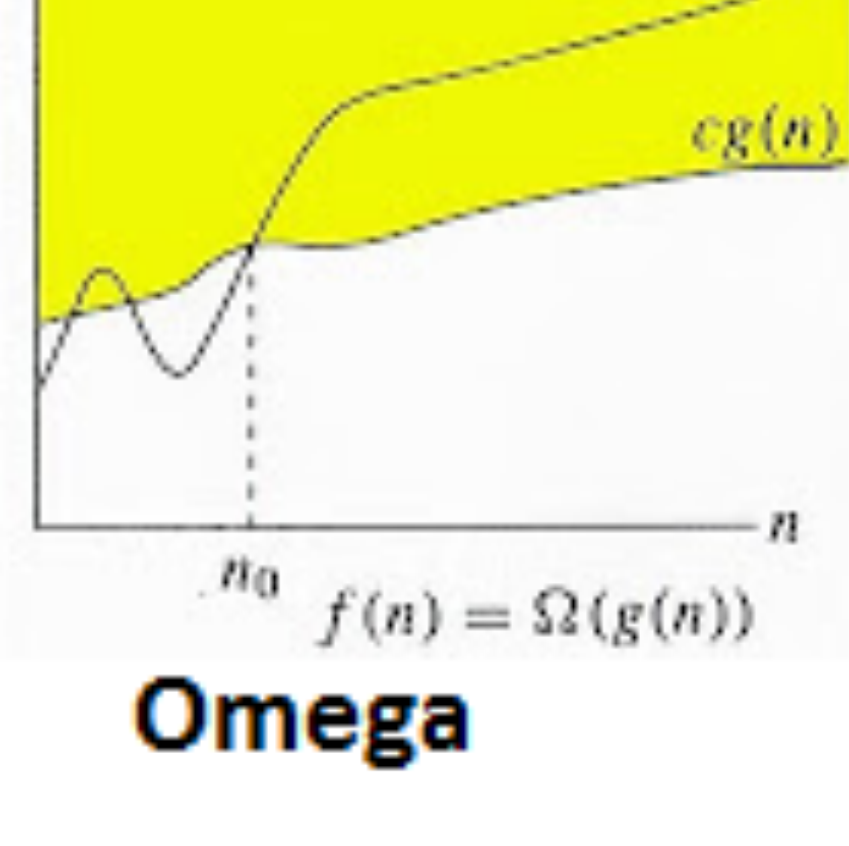
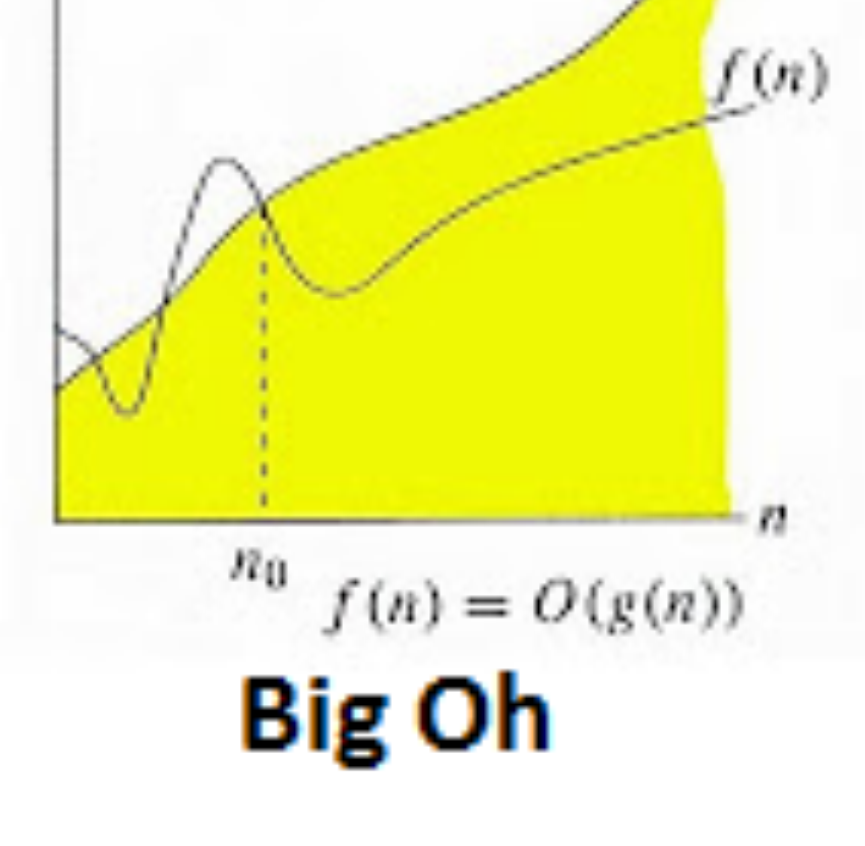
- ① Worst case
- ② Best case
- ③ Avg case

$$f(N) \leq c \cdot g(N) \quad N \geq N_0$$

$$f(N) \rightarrow \underline{O(g(N))}$$

Big-Oh defined

- Big-Oh is about finding an asymptotic upper bound.
- Formal definition of Big-Oh: $f(N) = O(g(N))$, if there exists positive constants c, N_0 such that $f(N) \leq c \cdot g(N)$ for all $N \geq N_0$.
- We are concerned with how f grows when N is large.
 - not concerned with small N or constant factors
- Lingo: " $f(N)$ grows no faster than $g(N)$."



$$T(N) = 2N^2 + 5N + 7 \quad O(N^2)$$

$$T(N) = 2N^2 + 2^N \log(N) + 2^N + 7$$

$$2^N \log(N)$$

$$(100)^3 \rightarrow 10^6$$

$$2^{100}$$

```
public static int Search(int[] arr, int item) {
    for (int i = 0; i < arr.length; i++) {
        if (arr[i] == item) {
            return i;
        }
    }
    return -1;
}
```

$$i=0 \quad i=1 \quad i=2 \quad i=3 \quad \dots \quad i=n-1$$

$$T(N) = N + 1 \quad \checkmark \quad O(N)$$

$$T(N) = N \quad \checkmark$$

```
public static int Maximum(int[] arr) {
    int max = arr[0];
    for (int i = 1; i < arr.length; i++) {
        if (arr[i] > max) {
            max = arr[i];
        }
    }
    return max;
}
```

$$i=1 \quad i=2 \quad i=3 \quad \dots \quad i=n-1$$

$$T(N) = n-1 \quad \checkmark \quad O(N)$$

$$T(N) = n+1 \quad \checkmark$$

```
public static int Search(int[] arr, int item) {
    int lo = 0;
    int hi = arr.length - 1;
    while (lo <= hi) {
        int mid = (lo + hi) / 2;
        if (arr[mid] == item) {
            return mid;
        } else if (arr[mid] > item) {
            hi = mid - 1;
        } else {
            lo = mid + 1;
        }
    }
    return -1;
}
```

$$N \rightarrow 1 \quad T = O(K)$$

$$\frac{N}{2} \rightarrow 1$$

$$\frac{N}{2^2} \rightarrow 1 \quad \frac{N}{2^k} = 1$$

$$\frac{N}{2^3} \rightarrow 1 \quad 2^k = N$$

$$\frac{N}{2^4} \rightarrow 1 \quad k = \log_2 N$$

$$\frac{N}{2^k} \rightarrow 1$$

$$\left\lceil \frac{N}{2^k} \right\rceil \rightarrow 1$$

$$2^k = N$$

```
while (i <= n) {
    System.out.println("Hey");
    i *= 2;
}
```

$$while(N > 0) \{$$

$$N = N/2$$

$$\log_2 x = y \quad x = 2^y$$

$$K \rightarrow \log_2(N)$$

$$\log_2 N$$

```
while (i <= n) {
    System.out.println("Hey");
    i += 2;
    i += 3;
}
```

$$i=0 \quad i=5 \quad i=10$$

$$1 \quad 1 \quad 1$$

```
for (i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        System.out.println("hey");
    }
}
```

$$n \times n \times 1 = n^2$$

$$i=1 \quad i=2 \quad i=3 \quad \dots \quad i=N$$

$$N^2$$

```
for (i = 1; i * i <= n; i++) {
    System.out.println("hey");
}
```

$$i^2 \leq N$$

$$i \leq \sqrt{N}$$

$$i=1$$

```
for (i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        for (k = 1; k <= n / 2; k++) {
            System.out.println("hey");
        }
    }
}
```

$$i=1 \quad i=2 \quad i=3 \quad i=4 \quad \dots \quad i=N$$

$$N \quad 2N \quad 3N \quad 4N \quad \dots \quad N^2$$

$$N(1+2+3+4+\dots+N)$$

$$N \times \left(\frac{N(N+1)}{2} \right) \quad O(N^3)$$

```
for (i = 1; i <= n; i++) {
    for (int j = 1; j <= i * i; j++) {
        for (int k = 1; k <= n / 2; k++) {
            System.out.println("hey");
        }
    }
}
```

$$i=1 \quad i=2 \quad i=3 \quad i=4 \quad \dots \quad i=N$$

$$\frac{N}{2} \quad 2 \times \frac{N}{2} \quad 3^2 \times \frac{N}{2} \quad 4^2 \times \frac{N}{2} \quad \dots \quad N^2 \times \frac{N}{2}$$

$$\frac{N}{2} (1^2 + 2^2 + 3^2 + 4^2 + \dots + N^2)$$

$$\frac{N}{2} \left(\frac{n(n+1)(2n+1)}{6} \right) = N^4$$

```
for (i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        for (int k = 1; k <= i * j; k++) {
            System.out.println("hey");
        }
    }
}
```

$$O(N^2) \quad \sin(x) > \cos(x)$$

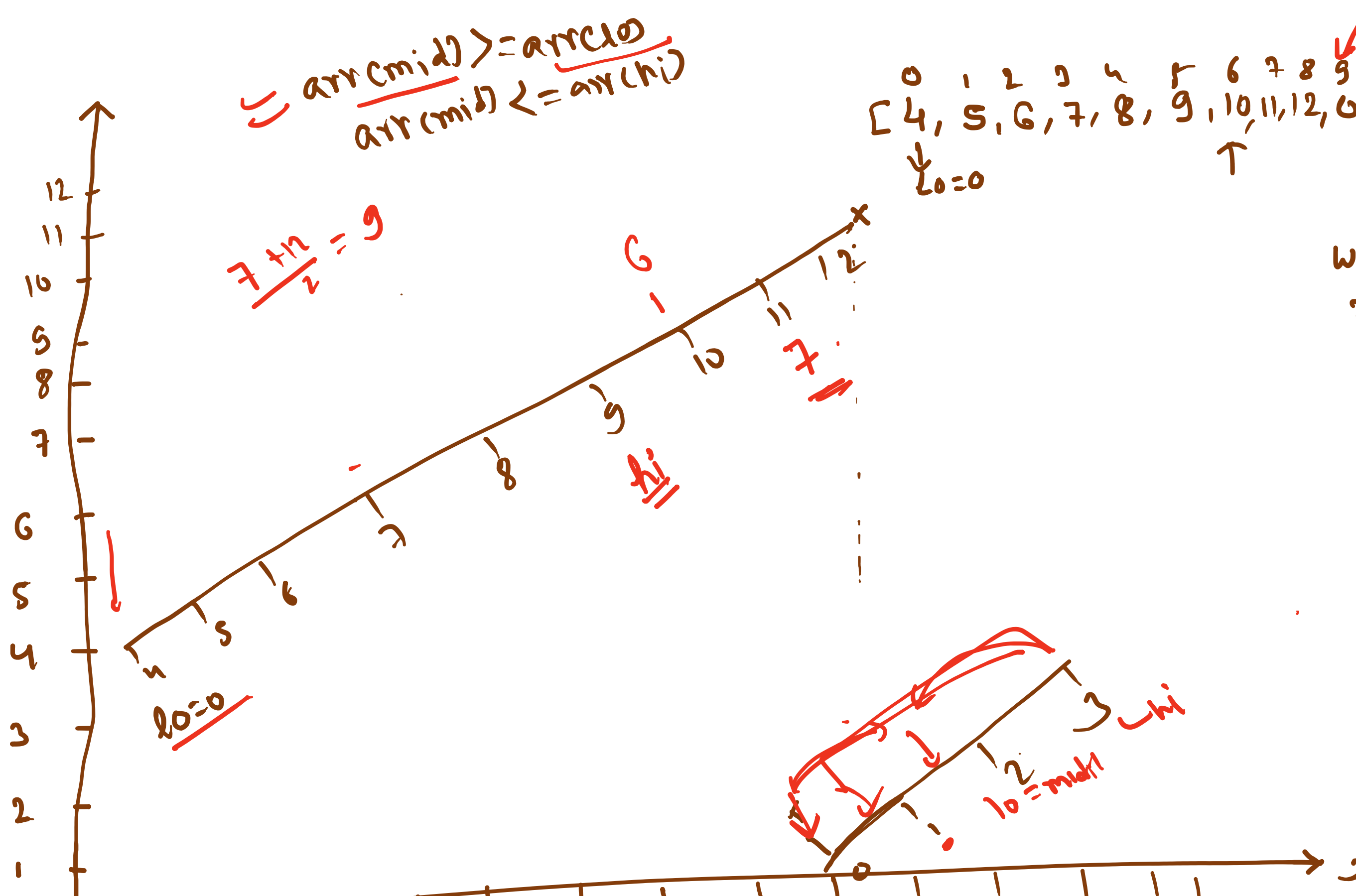
$$\log(x) > \cos(x)$$

$$i=1 \quad i=2 \quad i=3 \quad i=4 \quad \dots \quad i=N$$

$$N \quad \frac{N}{2} \quad \frac{N}{3} \quad \frac{N}{4} \quad \dots \quad \frac{N}{N}$$

$$N \left[1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{N} \right]$$

$$N \log(N)$$



$$zero = 0$$

$$two = arr.length - 1$$

$$[2, 1, 0, 0, 1, 2, 1, 2]$$

$$[2, 1, 0, 0, 1, 2, 1, 2]$$

$$[1, 1, 0, 0, 1, 2, 2, 2]$$

$$[0, 1, 0, 1, 2, 2, 2, 2]$$

$$[0, 0, 1, 1, 2, 2, 2, 2]$$

$$while (i <= two) \{$$

$$if (arr[mid] == 0) \{$$

$$swap(i, zero);$$

$$zero++;$$

$$i++;$$

$$\}$$

$$else if (arr[mid] == 2) \{$$

$$swap(i, two);$$

$$two--;$$

$$\}$$

$$}$$

$$}$$